

Use-Case Flow Specification

System: Clinical Trial Subject Recruitment Engine (Problem Statement #18)

Use Case ID: UC-001

Use Case Name: Complete Pre-Screening Survey (Includes: Evaluate Eligibility, Extends: Electronic Consent)

Primary Actor: Candidate

Secondary Actor: Receptionist (Assisted Intake)

Supporting Actors: SMS Notification Service, Clinical Researcher

Preconditions & Postconditions

- **Preconditions:**
 - The target clinical trial protocol is active, with inclusion/exclusion criteria configured by the Clinical Researcher.
 - The candidate is registered (or guided on-site by the Receptionist) with a valid mobile number and email.
- **Postconditions:**
 - **Success:** Subject is evaluated as "Eligible", the action is logged in the 21 CFR Part 11 audit trail, the e-Consent (eICF) workflow is unlocked, and an SMS confirmation is dispatched.
 - **Disqualified:** Candidate is marked "Ineligible", criteria failure codes are logged internally for the researcher, and a non-eligibility notice is displayed.

Main Success Scenario (Step-by-Step)

1. **Initiate Survey:** Candidate accesses the recruitment portal (or Receptionist launches assisted intake) and selects the relevant trial study from the catalogue.
2. **Complete Questionnaire:** Candidate inputs required demographic data, medical history, prior diagnoses, and current medications into structured survey fields.
3. **Submit & Validate:** Candidate submits responses; the system validates all mandatory fields and checks data formats in real time.
4. **<<includes>> Evaluate Eligibility:** System automatically executes the rule evaluation engine, comparing submitted metrics against active protocol inclusion/exclusion thresholds defined by the Clinical Researcher.
5. **Determine Qualification:** System confirms 100% adherence to inclusion criteria and zero disqualifying exclusion triggers; status is set to "Eligible - Pending Consent".
6. **Audit Trail Logging:** System generates a timestamped, cryptographically hashed 21 CFR Part 11 audit log entry recording the evaluation result.
7. **Dispatch Notification:** System triggers the **SMS Notification Service** to dispatch an automated confirmation text with a trial application reference ID.

8. <<extend>> **Electronic Consent:** System automatically opens the Electronic Informed Consent Form (eICF) workflow, allowing the candidate to review protocol disclosures and submit a legally binding digital signature.

Alternate & Exception Flows

- **AF-1: Disqualification on Exclusion Criteria (at Step 4–5)**
 - **4a.** The evaluation engine detects ≥ 1 disqualifying condition (e.g., conflicting medication, biomarker out of range).
 - **4b.** System flags candidate as Ineligible, logs failure codes internally for researcher review, and renders a standard non-eligibility message.
 - **4c.** Candidate is offered an opt-in option for a general registry for future trial notifications. Use case terminates.
- **AF-2: Electronic Consent Deferred by Candidate (at Step 8)**
 - **8a.** Candidate opts to review trial disclosures later or consult their primary physician.
 - **8b.** System saves pre-screened status as “Eligible - Consent Pending” with a 72-hour expiration window.
 - **8c.** SMS Notification Service sends a secure resume-link and reminder notification. Use case pauses until re-engagement.
- **EF-1: Network / Validation Service Interruption (at Step 3)**
 - **3a.** System detects network dropout or eligibility rule API timeout during submission.
 - **3b.** System caches survey responses locally, displays a retry dialog, and automatically syncs upon reconnection without data loss.